

have a good understanding of the design functionality.

Finally, a bitstream is generated, which has to be downloaded and designed into the FPGA for circuit verification.

ASIC design flow

The design process of ASIC also begins with functional specifications, which move onto the HDL code. Behavioural simulation is done using Questa sim, Model sim, and such other simulators. Before proceeding to synthesis, there is a step called DFT Insertion where flip-flop connections are checked for any faults. After synthesis, the static time analysis (STA) needs to be performed for which a specific STA engine is required. Therefore, the setup time and hold time delay need to be removed. However, their removal is quite challenging since they work at a very high frequency. (ASICs generally work at a very high frequency as compared to FPGAs.)

Next is equivalency checking and floor planning, which is related to standard cell libraries in an ASIC. Here, the HDL is converted into RTL. Floor planning is the step where the standard cell is replaced from the logical library onto the chip.

After that the design is given to the foundry. But before doing so it must be ensured that there are no mistakes or bugs in the design, else that will result in a huge loss. Initially, the foundry produces about 100 chips for basic testing. If a bug is found in the testing process, the entire process has to be repeated from the synthesis and floor planning stage until all the bugs have been removed from the ASIC. This takes a lot of time and, as a result, the time to market increases. This is the reason why FPGA is preferred for design prototyping.

After a wait of one to three months for the bug-free ASIC chips to arrive from the foundry, the next step is of place & route where STA is again performed to make wire connections (related to input-output).

Then comes the verification of second and third order effects. Once the faults have been rectified, the circuit needs to be verified. This is done by putting the ASIC on the board

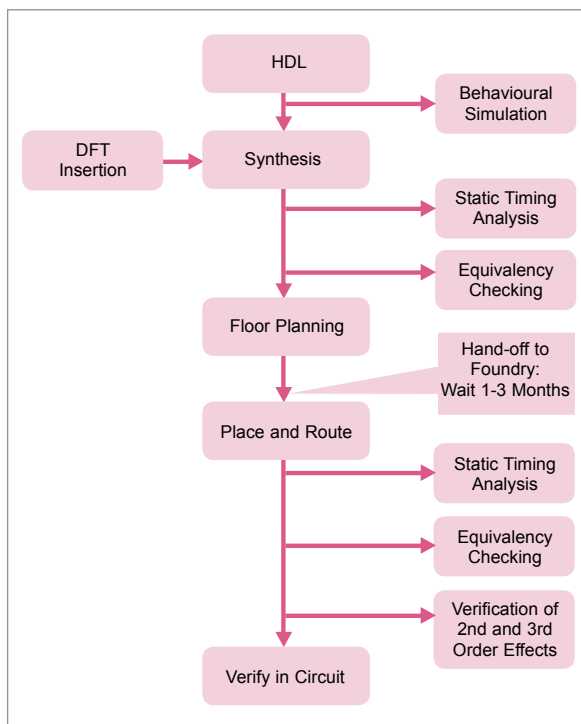


Fig. 2: Design flow of ASIC

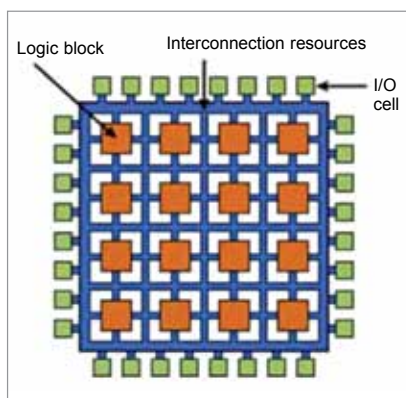


Fig. 3: Inside of an FPGA

along with the other components. After confirmation that the ASIC's operation meets expectations, it is released into the market.

So, ASIC has a lengthy design flow and takes a lot of time to manufacture, whereas FPGA takes much lesser time.

FPGA structure and its accelerated flow

Fig. 3 shows the inside on an FPGA, where the green coloured boxes are known as I/O Cells. The orange coloured boxes, called Logic Blocks, consist of configurable logic blocks (CLBs) and look up tables (LUTs). Depending on the FPGA, the I/O cells

and the logic blocks can change. The blue coloured lines represent the programmable interconnect resources, which allow the generation of different bit files.

If a bug is detected while verifying an FPGA circuit, a change is made in the HDL code to correct it, while the behavioural simulation is checked for any broken flow. Then the bit file is generated and later tested on the board to check its working. Generally, 3-4 days are required to correct a bug in FPGAs. But in ASICs that increases up to 6-7 months.

Once the ASIC RTL is designed, it is proto-

typed onto an FPGA to verify the design functionality. The benefit is that, with the help of RTL, bugs can be resolved easily along with any CDC or STA violations. However, it must be noted that there is no guarantee that the code result will be 100% bug-free. Only 60-80% of bugs can be resolved in the FPGA. The rest 40-20% of bugs have to be resolved in ASIC only.

FPGA can only accelerate the detection of bugs, which would take a lot of time to get resolved in ASIC. After detecting the bugs, the RTL is again changed from an FPGA to ASIC.

Peripheral connections

From Fig. 4 we can see that the chipset, CPU, and various other peripherals are externally connected to an FPGA because FPGA does not provide the facility to embed the CPU within itself (although many companies like Xilinx are working on changing that). This scenario, however, is absent in an ASIC, where the CPU can be embedded directly inside the ASIC.

Design challenges of ASIC

One of the major challenges faced in designing ASIC is clock routing (CLK-routing). In ASIC, the clock should